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Course: Calculus Beta

Assignment: MyLab Math - Uge 11

Solve the given equation by the method of undetermined coefficients.

$$y'' + 121y = 6 \cos x$$

The general solution to a nonhomogeneous differential equation $ay'' + by' + cy = G(x)$ is the sum of y_c and y_p , where y_c is the general solution to the complementary homogeneous equation $ay'' + by' + cy = 0$ and y_p is any particular solution to the nonhomogeneous equation.

$$y = y_c + y_p$$

First, find y_c by solving the homogeneous differential equation $y'' + 121y = 0$. The auxiliary equation for $y'' + 121y = 0$ is shown below.

$$r^2 + 121 = 0$$

Solve the auxiliary equation.

$$r^2 = -121$$

$$r = \pm 11i$$

The general solution to a second-order homogeneous differential equation whose auxiliary equation has two complex roots is $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$.

So, the general solution to $y'' + 121y = 0$ is $y = c_1 \cos 11x + c_2 \sin 11x$.

Now, find y_p , a particular solution to the given nonhomogeneous equation. The given equation has the form $ay'' + by' + cy = G(x)$.

When $G(x)$ is of the form $A \sin kx$ or $A \cos kx$, y_p has the form $B \cos kx + C \sin kx$. To find the coefficients of y_p , substitute y_p and its derivatives into the given nonhomogeneous differential equation and solve for the unknown coefficients B and C .

Note that in the given $G(x)$, k is 1. Find the first and second derivatives of $y_p = B \cos x + C \sin x$.

$$y_p' = -B \sin x + C \cos x \qquad y_p'' = -B \cos x - C \sin x$$

Substitute the expressions for y_p'' and y_p into the given nonhomogeneous equation.

$$y'' + 121y = 6 \cos x$$

$$-B \cos x - C \sin x + 121(B \cos x + C \sin x) = 6 \cos x$$

Combine like terms.

$$(121B - B) \cos x + (121C - C) \sin x = 6 \cos x$$

$$(120B) \cos x + (120C) \sin x = 6 \cos x$$

Compare the left and right sides of the resulting equation. The coefficient of $\cos x$ on the left side must be equal to the coefficient of $\cos x$ on the right side. Likewise, the coefficient of $\sin x$ on the left side must be equal to the coefficient of $\sin x$ on the right side.

$$120B = 6 \qquad 120C = 0$$

Solve for B and C .

$$B = \frac{1}{20} \qquad C = 0$$

Substitute these values into the expression for y_p and add the result to y_c to determine the solution to $y'' + 121y = 6 \cos x$.

$$y = y_c + y_p$$

$$y = c_1 \cos 11x + c_2 \sin 11x + \frac{1}{20} \cos x$$